

COPO

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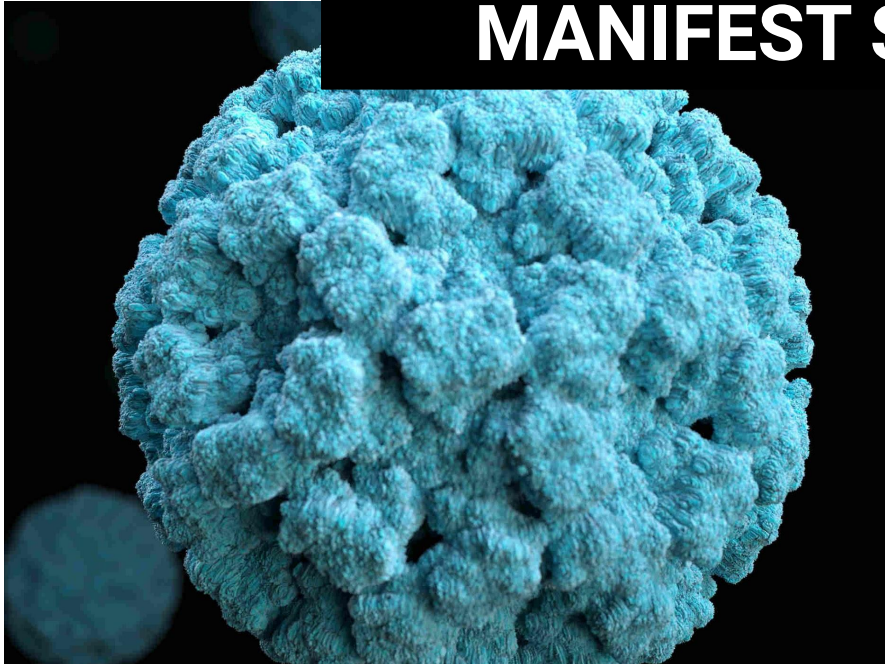


E Earlham
Institute

Decoding Living Systems

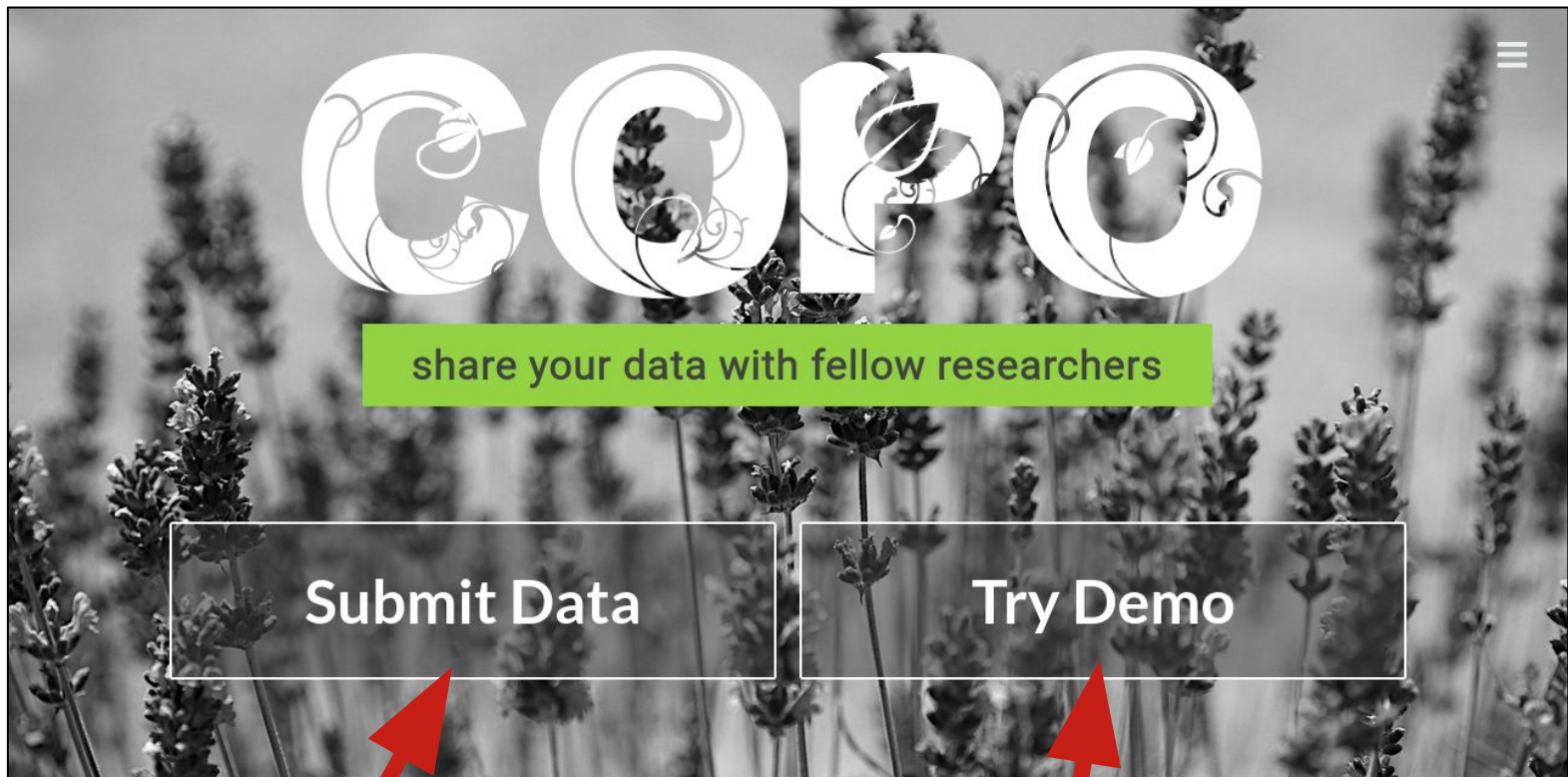


VISUAL DOCUMENTATION FOR MANIFEST SUBMISSION



What you need before starting:

- an ORCID account ([here](#) to register)
- contact El.COPO@earlham.ac.uk to be added to a manifest group
- make sure you have the latest version of the manifest ([here](#))
 - if you are submitting as part of the ASG project, please refer to the latest metadata manifest
 - if you are submitting as part of an ERGA project, please refer to the latest metadata manifest [here](#)



Once you are confident

To familiarise with the platform

IMPORTANT - Submissions to ENA through the demo server will not persist, accessions created will be deleted after 24 hours.

We also update this server frequently, so it may occasionally be unreachable and you may find we have deleted your past uploads.



Earlham Institute

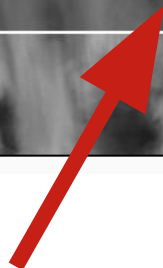
Log in with your ORCID credentials:



share your data with fellow researchers

Submit Data

Try Demo



Create a profile:

The screenshot displays the COPO Work Profiles interface. At the top, the COPO logo is on the left, and a notification bell, a document icon, and a user profile 'alice' are on the right. Below the logo, the text 'Work Profiles' is followed by an orange lightbulb icon and a blue circle with a white plus sign. A red arrow points to this plus sign. Below these icons are buttons for 'Select all', 'Select none', '+ Add', and 'Edit'. A search bar labeled 'Search Work Profiles' is on the right. The main content area shows 'Showing 1 to 1 of 1 entries' and a profile card for 'dtol (DTOL)'. The card includes the text 'Created: Mon, 26 Oct 2020 17:38' and 'Description: test'. At the bottom of the card are six component buttons: 'Samples' (0), 'Datafiles' (0), 'Submissions' (0), 'Publications' (0), 'Metadata Template' (0), and 'People' (1). On the right side, there is a sidebar with 'Info' and 'Help' tabs. The 'Help' tab is active, showing a 'Search Help' bar and a list of 'Help Topics' with five items, each preceded by a plus sign. At the bottom of the sidebar, it says '1 to 5 of 5 topics' and has 'Previous', '1', and 'Next' buttons.

COPO

Work Profiles

Select all Select none + Add Edit

Search Work Profiles

Showing 1 to 1 of 1 entries

dtol (DTOL)

Created:
Mon, 26 Oct 2020 17:38

Description:
test

Samples 0 Datafiles 0 Submissions 0 Publications 0 Metadata Template 0 People 1

Show 10 entries

Info Help

Search Help

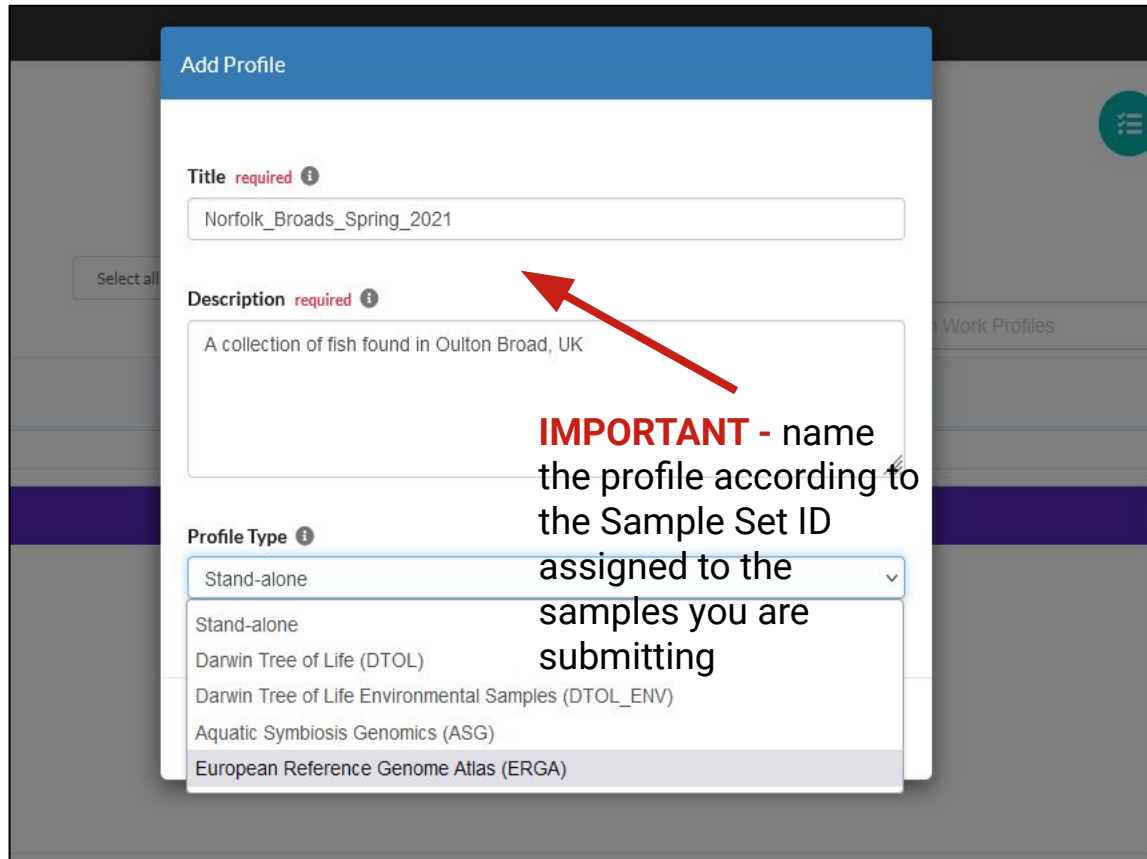
Help Topics

- + What is a COPO profile?
- + How do I create a profile?
- + How many profiles can I have in COPO?
- + What is a profile component?
- + How do I create components within a profile?

1 to 5 of 5 topics

Previous 1 Next

Create a profile:



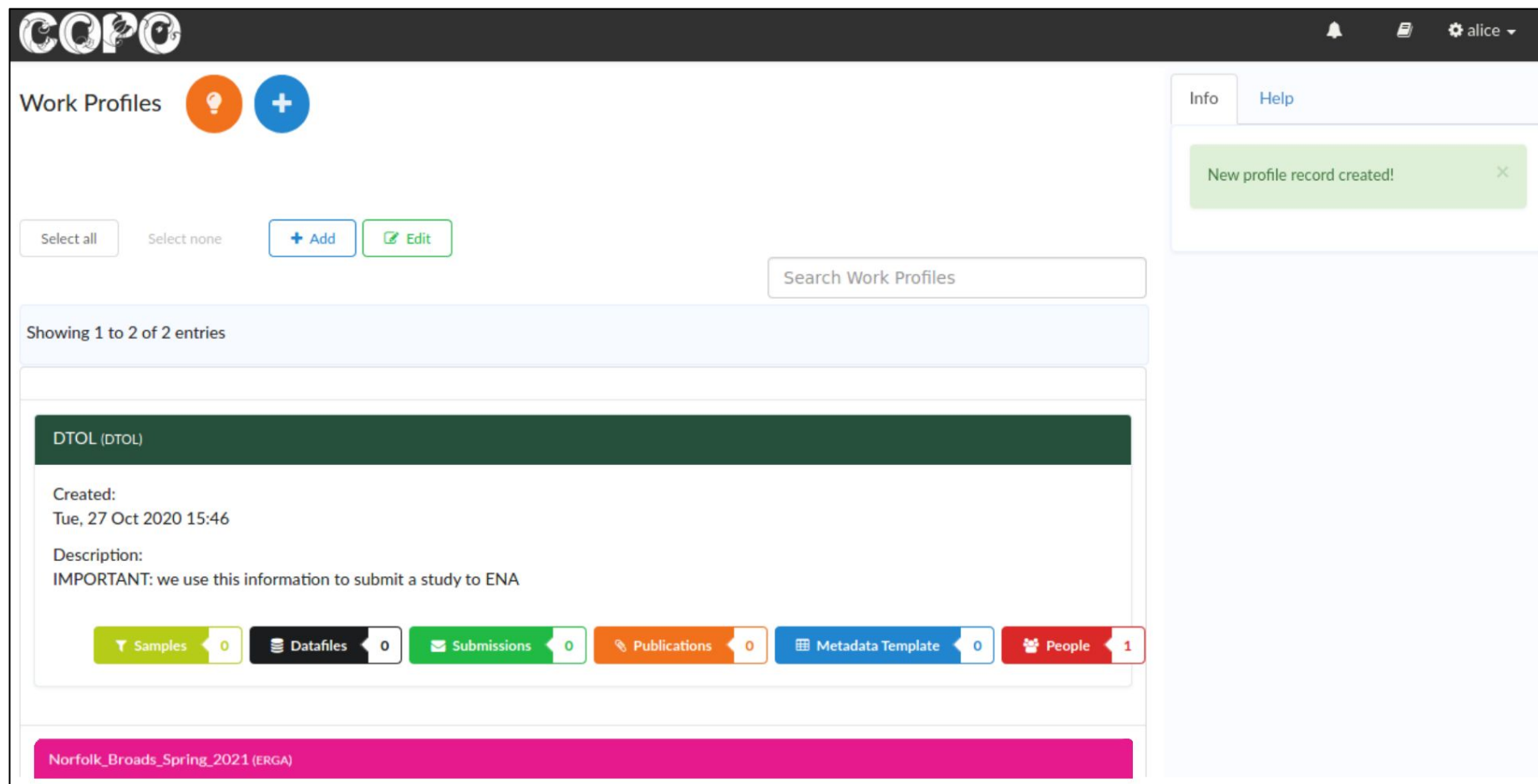
The screenshot shows a web form titled "Add Profile". It has three main sections: "Title", "Description", and "Profile Type". The "Title" field contains "Norfolk_Broads_Spring_2021". The "Description" field contains "A collection of fish found in Oulton Broad, UK". The "Profile Type" dropdown menu is open, showing options: "Stand-alone", "Darwin Tree of Life (DTOL)", "Darwin Tree of Life Environmental Samples (DTOL_ENV)", "Aquatic Symbiosis Genomics (ASG)", and "European Reference Genome Atlas (ERGA)". A red arrow points to the "Description" field. A red text box with white text is overlaid on the right side of the form, stating: "IMPORTANT - name the profile according to the Sample Set ID assigned to the samples you are submitting".

IMPORTANT - name the profile according to the Sample Set ID assigned to the samples you are submitting

IMPORTANT - If you cannot see the project that you are associated with in the "Profile Type" drop down menu, you need to be added to the relevant group.

Please stop here and get in touch with EI.COPO@earlham.ac.uk

This is your new profile:



The screenshot displays the COPPO Work Profiles interface. At the top, the COPPO logo is on the left, and a dark header bar contains a bell icon, a document icon, and a user profile labeled 'alice'. Below the header, the 'Work Profiles' section features a lightbulb icon and a blue circle with a white plus sign. A toolbar includes 'Select all', 'Select none', '+ Add', and 'Edit' buttons. A search bar labeled 'Search Work Profiles' is positioned to the right. A notification box on the right side states 'New profile record created!'. The main content area shows 'Showing 1 to 2 of 2 entries'. The first entry, 'DTOL (DTOL)', is highlighted in dark green. It includes a 'Created' timestamp of 'Tue, 27 Oct 2020 15:46' and a 'Description' stating 'IMPORTANT: we use this information to submit a study to ENA'. Below the description is a horizontal bar with six colored buttons: 'Samples' (yellow, 0), 'Datafiles' (black, 0), 'Submissions' (green, 0), 'Publications' (orange, 0), 'Metadata Template' (blue, 0), and 'People' (red, 1). The second entry, 'Norfolk_Broads_Spring_2021 (ERGA)', is partially visible at the bottom in a pink bar.

COPPO

Work Profiles

Select all Select none + Add Edit

Search Work Profiles

Showing 1 to 2 of 2 entries

DTOL (DTOL)

Created:
Tue, 27 Oct 2020 15:46

Description:
IMPORTANT: we use this information to submit a study to ENA

Samples 0 Datafiles 0 Submissions 0 Publications 0 Metadata Template 0 People 1

Norfolk_Broads_Spring_2021 (ERGA)

Info Help

New profile record created!

Upload a manifest:

Showing 1 to 4 of 4 entries 1 row selected

Norfolk_Broads_Spring_2021 (ERGA)

Created:
Mon, 05 Dec 2022 15:04

Description:
A collection of fish collected at Oulton Broad, UK

Actions ▶

Submit ERGA Manifest

Components ▶



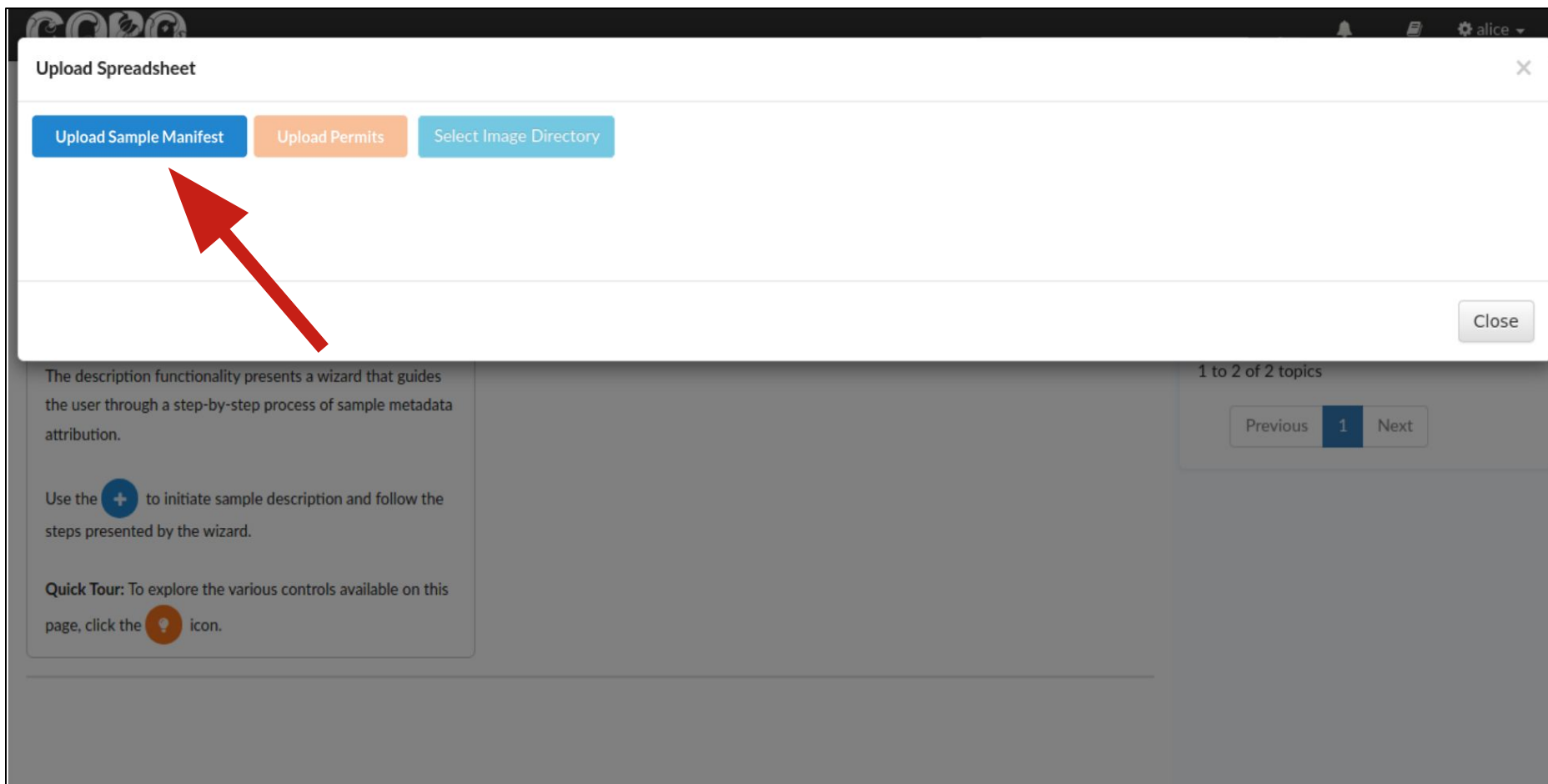
The colour and wording of this panel will differ depending on which project you are submitting a manifest for.

Upload a manifest:

The screenshot displays the CPOPO web application interface. At the top left is the CPOPO logo. Below it, the text 'Profile: Profile 1' and 'Samples' are visible. A dark header bar contains a notification bell, a document icon, and a user profile 'alice'. A toolbar with four icons (database, email, link, and group) is positioned below the header. A 'Samples - Getting started' sidebar on the left provides instructions on using the platform. The main content area features three circular buttons: an orange lightbulb icon, a green grid icon, and a teal list icon. A red arrow points to the green grid icon, which is highlighted by a tooltip that reads 'Add Sample(s) from Profile 1 Spreadsheet'. On the right, a 'Help' sidebar shows a search bar and a list of help topics: 'Describing samples' and 'Sample types'.

*Only sample managers will see the third (teal) button next to the green button, if this is not your role do not worry about that, you would not need it

Upload a manifest:



The screenshot shows a web application interface. At the top, there is a dark header bar with social media icons on the left and a user profile 'alice' on the right. Below the header, a white modal dialog box titled 'Upload Spreadsheet' is open. It contains three buttons: 'Upload Sample Manifest' (blue), 'Upload Permits' (orange), and 'Select Image Directory' (light blue). A large red arrow points to the 'Upload Sample Manifest' button. In the bottom right corner of the dialog is a 'Close' button. The background of the application is dimmed and shows a sidebar with text about a description wizard and a 'Quick Tour' section, and a main content area with a pagination control showing '1 to 2 of 2 topics' and buttons for 'Previous', '1', and 'Next'.

Upload Spreadsheet

Upload Sample Manifest Upload Permits Select Image Directory

Close

The description functionality presents a wizard that guides the user through a step-by-step process of sample metadata attribution.

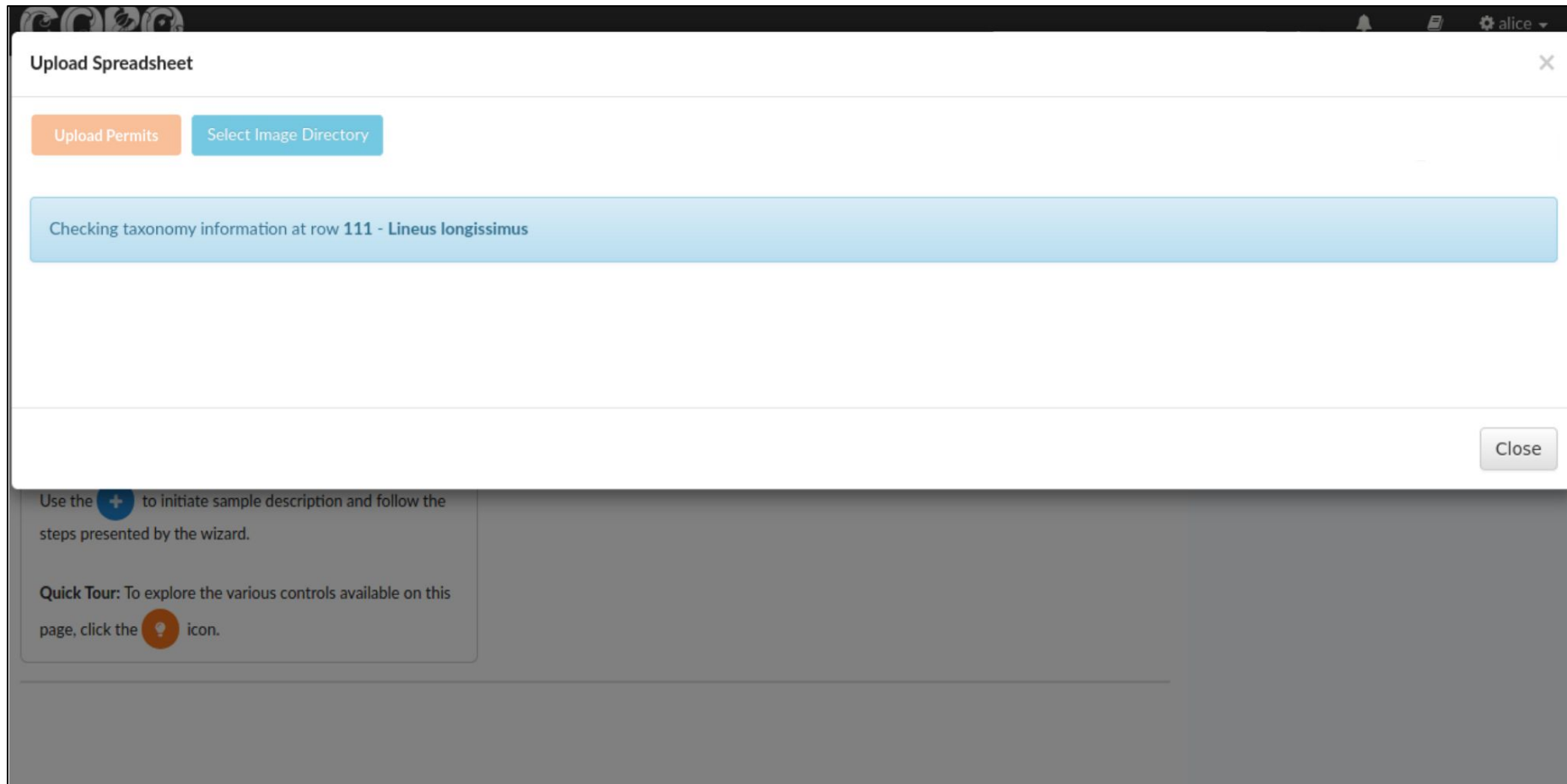
Use the  to initiate sample description and follow the steps presented by the wizard.

Quick Tour: To explore the various controls available on this page, click the  icon.

1 to 2 of 2 topics

Previous 1 Next

COPPO will validate taxonomy information and manifest format:



What's happening (part 1):

- COPO is retrieving information from NCBI (National Center for Biotechnology Information) to check that the taxonomy information provided is correct. We also check that the TAXON_ID is submittable to ENA.
- If some of the taxonomy fields are missing (excluding SCIENTIFIC_NAME), COPO will make an educated guess about what these fields should be and will display warnings about what is going to be filled in.

The taxonomy checks will be much quicker if you fill in TAXON_ID.

- COPO will substitute scientific name synonyms with the official scientific name. It will display a warning and also record the change in OTHER_INFORMATION
- If any taxonomy field is incorrect, COPO will display an error and reject the manifest. You can download the error list with the red “Export Errors” button on the top right.
- **If you are unhappy with any of the warning/errors, DO NOT press finish; instead contact NCBI to discuss the taxonomy**

Manifest uploaded -taxonomy warnings:-

alice

Upload Spreadsheet

Upload Permits

Select Image Directory

Export Errors

Synonym warning: *Asterina gibbosa* at row 294 is a synonym of *Asterina gibbosa* (Pennant, 1777). COPO will substitute the official scientific name.

Synonym warning: *Asterina gibbosa* at row 295 is a synonym of *Asterina gibbosa* (Pennant, 1777). COPO will substitute the official scientific name.

Synonym warning: *Asterina gibbosa* at row 296 is a synonym of *Asterina gibbosa* (Pennant, 1777). COPO will substitute the official scientific name.

Synonym warning: *Asterina gibbosa* at row 297 is a synonym of *Asterina gibbosa* (Pennant, 1777). COPO will substitute the official scientific name.

Synonym warning: *Aglais io* at row 336 is a synonym of *Inachis io*. COPO will substitute the official scientific name.

Synonym warning: *Aglais io* at row 337 is a synonym of *Inachis io*. COPO will substitute the official scientific name.

Synonym warning: *Aglais io* at row 338 is a synonym of *Inachis io*. COPO will substitute the official scientific name.

Synonym warning: *Satyrrium walbum* at row 359 is a synonym of *Satyrrium w-album*. COPO will substitute the official scientific name.

Synonym warning: *Aglais io* at row 385 is a synonym of *Inachis io*. COPO will substitute the official scientific name.

Synonym warning: *Melitaea athalia* at row 450 is a synonym of *Mellicta athalia*. COPO will substitute the official scientific name.

Synonym warning: *Colias crocea* at row 498 is a synonym of *Colias croceus*. COPO will substitute the official scientific name.

Synonym warning: *Colias crocea* at row 499 is a synonym of *Colias croceus*. COPO will substitute the official scientific name.

Synonym warning: *Colias crocea* at row 500 is a synonym of *Colias croceus*. COPO will substitute the official scientific name.

Synonym warning: *Colias crocea* at row 501 is a synonym of *Colias croceus*. COPO will substitute the official scientific name.

Synonym warning: *Colias crocea* at row 502 is a synonym of *Colias croceus*. COPO will substitute the official scientific name.

Synonym warning: *Colias crocea* at row 503 is a synonym of *Colias croceus*. COPO will substitute the official scientific name.

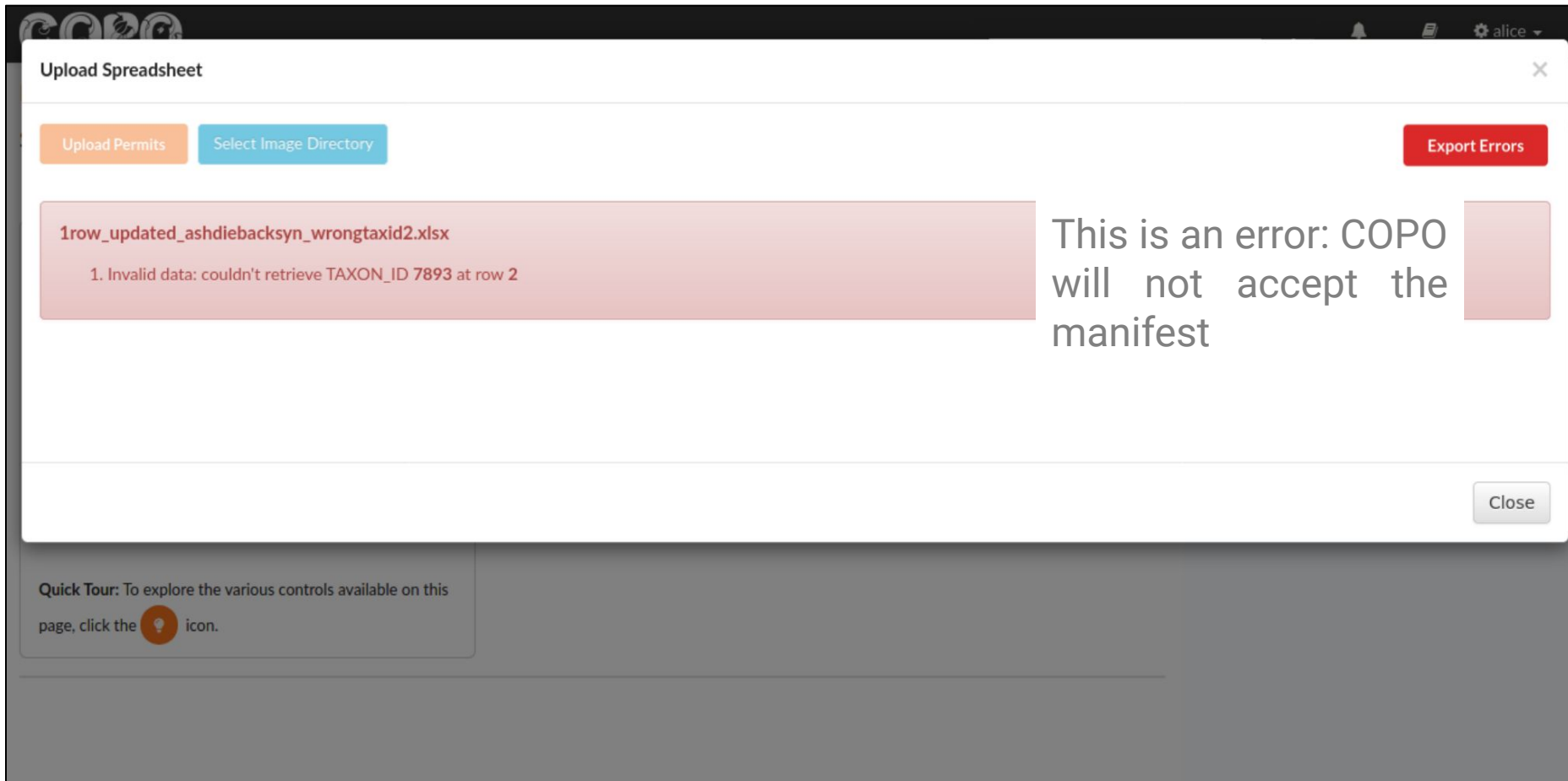
Synonym warning: *Colias crocea* at row 504 is a synonym of *Colias croceus*. COPO will substitute the official scientific name.

Synonym warning: *Melitaea athalia* at row 516 is a synonym of *Mellicta athalia*. COPO will substitute the official scientific name.

This is a warning:
COPO will make
these changes

SAN.xlsx

Manifest uploaded -taxonomy errors-:



The screenshot shows a web application interface with a dark header bar containing navigation icons and a user profile 'alice'. A white modal window titled 'Upload Spreadsheet' is open. It features two buttons: 'Upload Permits' (orange) and 'Select Image Directory' (blue). In the top right corner of the modal is a red 'Export Errors' button. A red error message box displays the filename '1row_updated_ashdiebacksyn_wrongtaxid2.xlsx' and the message '1. Invalid data: couldn't retrieve TAXON_ID 7893 at row 2'. To the right of this box, the text 'This is an error: COPO will not accept the manifest' is displayed. A 'Close' button is located at the bottom right of the modal. Below the modal, a 'Quick Tour' box provides instructions on how to explore the controls.

Upload Spreadsheet

Upload Permits Select Image Directory Export Errors

1row_updated_ashdiebacksyn_wrongtaxid2.xlsx

1. Invalid data: couldn't retrieve TAXON_ID 7893 at row 2

This is an error: COPO will not accept the manifest

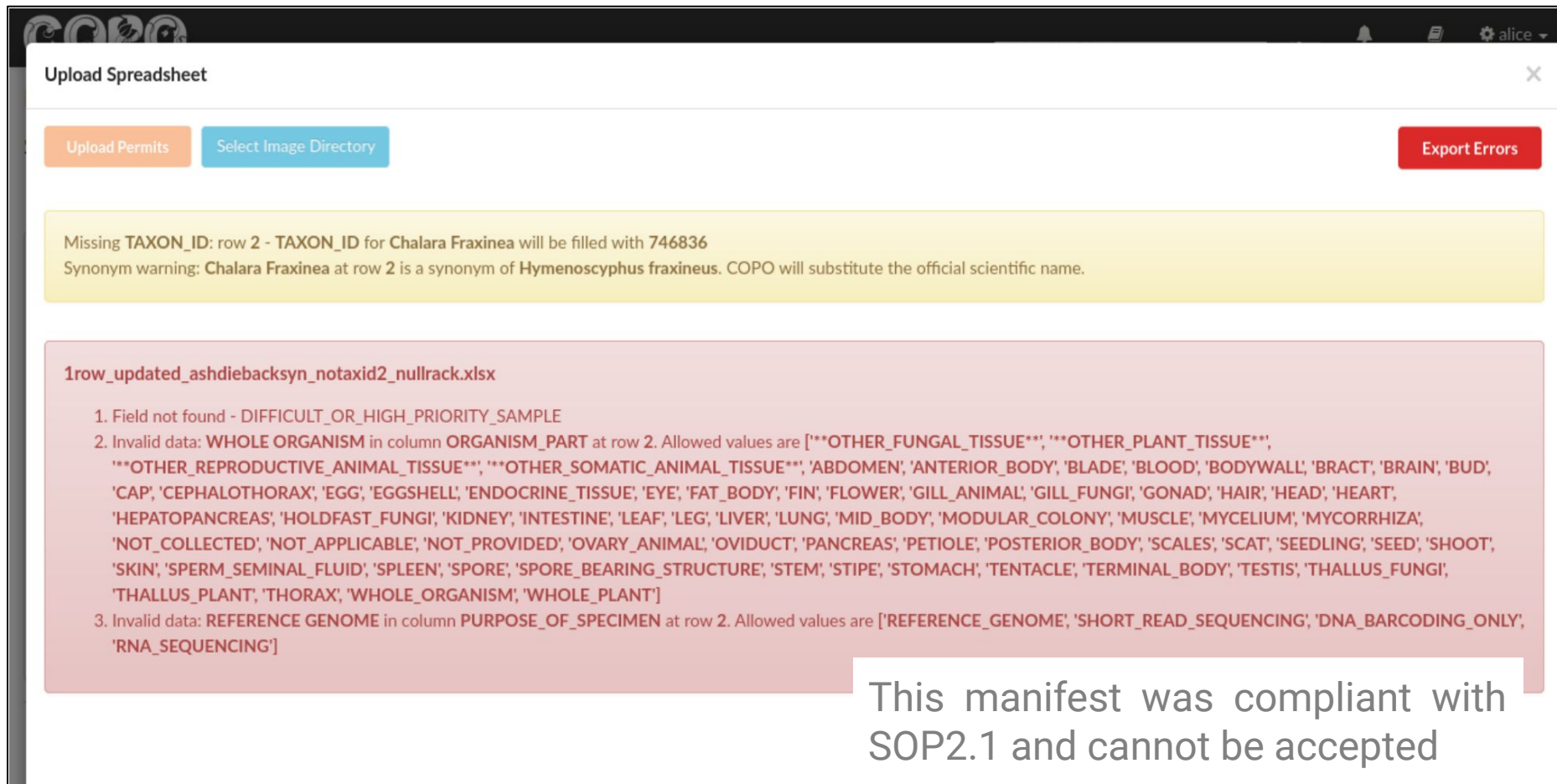
Close

Quick Tour: To explore the various controls available on this page, click the  icon.

What's happening (part 2):

- **ONLY IF** the taxonomy validation was successful and produced no errors, will COPO check the manifest against SOP specifications. This includes checks for general format, missing data, cells that have restricted values, etc..

Manifest uploaded -compliance errors-:



The screenshot shows a web interface titled "Upload Spreadsheet" with a close button (X) in the top right corner. Below the title bar, there are three buttons: "Upload Permits" (orange), "Select Image Directory" (blue), and "Export Errors" (red). A yellow warning box contains the following text:

Missing **TAXON_ID**: row 2 - **TAXON_ID** for *Chalara Fraxinea* will be filled with 746836
Synonym warning: *Chalara Fraxinea* at row 2 is a synonym of *Hymenoscyphus fraxineus*. COPO will substitute the official scientific name.

Below the warning box, a red box displays the filename "1row_updated_ashdiebacksyn_notaxid2_nullrack.xlsx" and a list of three errors:

1. Field not found - DIFFICULT_OR_HIGH_PRIORITY_SAMPLE
2. Invalid data: **WHOLE ORGANISM** in column **ORGANISM_PART** at row 2. Allowed values are ['**OTHER_FUNGAL_TISSUE**', '**OTHER_PLANT_TISSUE**', '**OTHER_REPRODUCTIVE_ANIMAL_TISSUE**', '**OTHER_SOMATIC_ANIMAL_TISSUE**', 'ABDOMEN', 'ANTERIOR_BODY', 'BLADE', 'BLOOD', 'BODYWALL', 'BRACT', 'BRAIN', 'BUD', 'CAP', 'CEPHALOTHORAX', 'EGG', 'EGGSHELL', 'ENDOCRINE_TISSUE', 'EYE', 'FAT_BODY', 'FIN', 'FLOWER', 'GILL_ANIMAL', 'GILL_FUNGI', 'GONAD', 'HAIR', 'HEAD', 'HEART', 'HEPATOPANCREAS', 'HOLDFAST_FUNGI', 'KIDNEY', 'INTESTINE', 'LEAF', 'LEG', 'LIVER', 'LUNG', 'MID_BODY', 'MODULAR_COLONY', 'MUSCLE', 'MYCELIUM', 'MYCORRHIZA', 'NOT_COLLECTED', 'NOT_APPLICABLE', 'NOT_PROVIDED', 'OVARY_ANIMAL', 'OVIDUCT', 'PANCREAS', 'PETIOLE', 'POSTERIOR_BODY', 'SCALES', 'SCAT', 'SEEDLING', 'SEED', 'SHOOT', 'SKIN', 'SPERM_SEMINAL_FLUID', 'SPLEEN', 'SPORE', 'SPORE_BEARING_STRUCTURE', 'STEM', 'STIPE', 'STOMACH', 'TENTACLE', 'TERMINAL_BODY', 'TESTIS', 'THALLUS_FUNGI', 'THALLUS_PLANT', 'THORAX', 'WHOLE_ORGANISM', 'WHOLE_PLANT']
3. Invalid data: **REFERENCE GENOME** in column **PURPOSE_OF_SPECIMEN** at row 2. Allowed values are ['REFERENCE_GENOME', 'SHORT_READ_SEQUENCING', 'DNA_BARCODING_ONLY', 'RNA_SEQUENCING']

A white text box at the bottom right of the red error box contains the text: "This manifest was compliant with SOP2.1 and cannot be accepted".

The errors should guide you to correct the manifest. You will have to correct and re-upload the updated spreadsheet

Manifest uploaded -correct manifest-:



Upload Spreadsheet 

Upload Permits

Select Image Directory

Finish

Missing TAXON_ID: row 2 - TAXON_ID for Chalara Fraxinea will be filled with 746836

Synonym warning: Chalara Fraxinea at row 2 is a synonym of Hymenoscyphus fraxineus. COPO will substitute the official scientific name.

Sample Metadata

Sample Images

Show

10

 entries

Search:

SERIES 	RACK_OR_PLATE_ID 	TUBE_OR_WELL_ID 	SPECIMEN_ID 	ORDER_OR_GROUP 	FAMILY 	GENUS 	TAXON_ID 	SCIENTIFIC_NAME
1	null	FF06763795	SAN0000100	HELOTIALES	HELOTIACEAE	HYMENOSCYPHUS	746836	HYMENOSCYPHUS FRAXINEUS

Check the warnings if present

Manifest uploaded -correct manifest-:



alice

Upload Spreadsheet

Upload Permits

Select Image Directory

Finish

Missing TAXON_ID: row 2 - TAXON_ID for *Chalara Fraxinea* will be filled with 746836

Synonym warning: *Chalara Fraxinea* at row 2 is a synonym of *Hymenoscyphus fraxineus*. COPO will substitute the official scientific name.

Sample Metadata

Sample Images

Preview of the manifest

Show 10 entries

Search:

SERIES	RACK_OR_PLATE_ID	TUBE_OR_WELL_ID	SPECIMEN_ID	ORDER_OR_GROUP	FAMILY	GENUS	TAXON_ID	SCIENTIFIC_NAME
1	null	FF06763795	SAN0000100	HELOTIALES	HELOTIACEAE	HYMENOSCYPHUS	746836	HYMENOSCYPHUS FRAXINEUS

Upload images:



Upload Spreadsheet

Upload Permits Select Image Directory Finish

Missing **TAXON_ID**: row 2 - **TAXON_ID** for *Chalara Fraxinea* will be filled with 746836
Synonym warning: *Chalara Fraxinea* at row 2 is a synonym of *Hymenoscyphus fraxineus*. COPO will substitute the official scientific name.

Sample Metadata Sample Images

Show entries

SERIES	RACK_OR_PLATE_ID	TUBE_OR_WELL_ID	SPECIMEN_ID	ORDER_OR_GROUP	FAMILY	GENUS	TAXON_ID	SCIENTIFIC_NAME
1	null	FF06763795	SAN0000100	HELOTIALES	HELOTIACEAE	HYMENOSCYPHUS	746836	HYMENOSCYPHUS FRAXINEUS

Search:

File names of any associated images must be the same as "SPECIMEN_ID" with the extension ".jpg" or ".png"

Upload images:

alice

Upload Spreadsheet

[Upload Permits](#) [Select Image Directory](#) [Finish](#)

Missing **TAXON_ID**: row 2 - **TAXON_ID** for **Chalara Fraxinea** will be filled with **746836**
Synonym warning: **Chalara Fraxinea** at row 2 is a synonym of **Hymenoscyphus fraxineus**. COPO will substitute the official scientific name.

[Sample Metadata](#) [Sample Images](#)

Show entries

Search:

Specimen ID	Image File	Image
SAN0000100	SAN0000100.png	

Showing 1 to 1 of 1 entries

[Previous](#) [1](#) [Next](#)

Create the samples in COPO by clicking “Finish”:

alice

Upload Spreadsheet

Upload Permits

Select Image Directory

Finish

Missing **TAXON_ID**: row 2 - **TAXON_ID** for *Chalara Fraxinea* will be filled with **746836**
Synonym warning: *Chalara Fraxinea* at row 2 is a synonym of *Hymenoscyphus fraxineus*. COPO will substitute the official scientific name.

Sample Metadata

Sample Images

Show

10

 entries

Search:

Specimen ID	Image File	Image
SAN0000100	SAN0000100.png	

Showing 1 to 1 of 1 entries

Previous

1

Next

Uploaded samples:



alice

Profile: Profile 1

Samples



Select all

Select filtered

Clear selection

Export CSV

Source list

Describe

Edit

Search Samples

Showing 1 to 1 of 1 records Click  beside a record to view extra details

	Name	Collected By	Date of Collection	Sex	Lifestage	Identified By	Identifier Affiliation	Common Name	Spec Id
	⇅	⇅	⇅	⇅		⇅	⇅	⇅	⇅
		JULIE AHRINGER RICHARD DURBIN	2020-08-22	MALE	SPORE_BEARING_STRUCTURE	MARK BLAXTER	SANGER INSTITUTE	lugworm	SANC

Info

Help

Search Help

Help Topics

 Describing samples

 Sample types

1 to 2 of 2 topics

Previous

1

Next

Sample Managers:

- At this point, the sample managers will be emailed that a new manifest has been uploaded. They will be able to accept or reject each sample after inspection.

After samples are approved by a sample manager, they will be submitted to ENA (European Nucleotide Archive). Accessions can be inspected by scrolling right in the previous view.

Accepted samples:



alice

Profile: Profile 1

Samples



Select all

Select filtered

Clear selection

Export CSV

Source list

Describe

Edit

Search Samples

Showing 1 to 1 of 1 records Click  beside a record to view extra details

Voucher ID	Manifest ID	Biosample Accession	SRA Accession	Submission Accession	Status	Public Name	Date Modified
NOT APPLICABLE	1943d3d2-7976-4563-859e-6877fb1d2560	SAMEA7491029	ERS5247681	ERA3031374	accepted	gyHymFrax1	[object Object]

Search Help

Help Topics

+ Describing samples

+ Sample types

1 to 2 of 2 topics

Previous

1

Next



If you encounter any errors or unexpected behaviour, please email EI.COPO@earlham.ac.uk or open a GitHub issue in the [COPO repository](#) at the bottom of the page.

If you have a feature request for future releases, please open an issue in the GitHub repository.

We are currently working on a pipeline to update uploaded samples. For the time being, if you need to update a sample please email us at EI.COPO@earlham.ac.uk specifying the biosample accession, the field to update and the new value. Likewise, if you need to update the taxonomy, please send the biosample accession and the new scientific name then, we will update all the taxonomy fields accordingly.





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Decoding Living Systems